

Progress Report

OFDM Modulation Scheme and Channel Estimation

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Abstract

This guide is made by us to develop an understanding of the OFDM modulation scheme and various Channel Estimation techniques, with a future goal of implementing the same on an SDR.

Keywords: Channel Estimation,, OFDM

1. Why not PSK?

In the previous report, we reviewed a Phase Shift Keying based receiver and transmitter, and implemented a version on SDR. We now present where PSK suffers, and how OFDM mitigates the shortcomings.

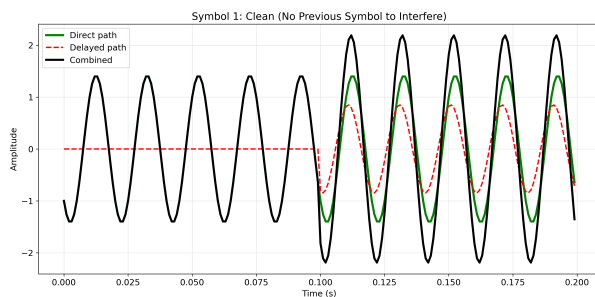


Figure 1. ISI in QPSK

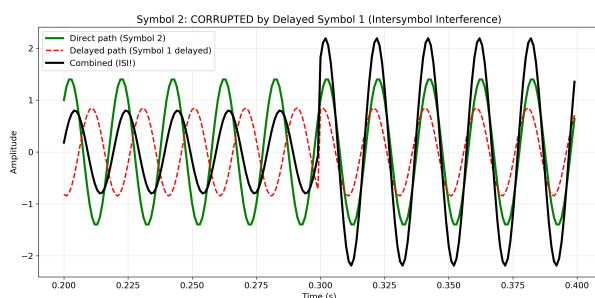


Figure 2. ISI in QPSK, pt2

The analysis thus far, has been assuming non-fading ($\alpha \approx 1$), AWGN channels. But in actual channels, There are reflected signals, free space path loss, scattering, and other non-idealities,

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which attenuate the signal, cause delayed copy superposition. This causes Inter Symbol Interference, shown in Figure 1. This effects the received constellation in terms of both rotation and gain.

1.1. What is OFDM and how does it combat ISI?

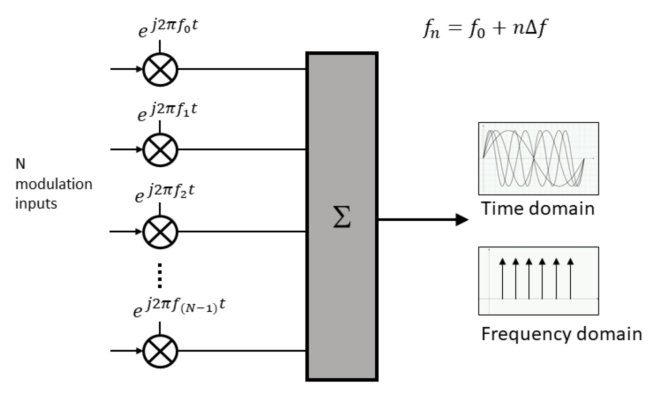


Figure 3. OFDM Modulation

OFDM (Orthogonal Frequency Division Multiplexing) is a scheme, which splits data of Length N, which would be transmitted at some rate R, to $\frac{N}{m}$, and is transmitted at a rate $\frac{R}{m}$. For example, in QPSK, a bitstream of the form

$$[1, 0, 0, 1, 1, 0, 1, 1]$$

is encoded as

$$[1, 0] [0, 1] [1, 0] [1, 1]$$

Which translates to

$$1 - j, -1 + j, 1 - j, -1 - j$$

And is transmitted with some symbol duration T_s . In OFDM, multiple symbols can be mapped onto different subcarriers in the frequency domain, effectively extending the overall symbol duration to $4T_s$. Unlike traditional FDM, which requires non-overlapping subcarriers with guard bands, OFDM allows subcarriers to overlap

in frequency, with peaks separated by $\Delta f = \frac{k}{4T_s}$ where $k \geq 1 \in N$. Consider a discrete-time OFDM system with N subcarriers. The transmitted OFDM symbol in the frequency domain is represented as:

$$\mathbf{X}[k] = [X_0[k], X_1[k], \dots, X_{N-1}[k]]^T \quad (1)$$

where $X_m[k]$ is the complex symbol transmitted on the m -th subcarrier during the k -th OFDM symbol period.

The time-domain OFDM signal is obtained through the Inverse Discrete Fourier Transform (IDFT):

$$x[n, k] = \frac{1}{\sqrt{N}} \sum_{m=0}^{N-1} X_m[k] e^{j2\pi mn/N}, \quad n = 0, 1, \dots, N-1 \quad (2)$$

The orthogonality condition between subcarriers is mathematically expressed as:

$$\int_0^{T_s} e^{j2\pi f_i t} e^{-j2\pi f_j t} dt = \begin{cases} T_s & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (3)$$

where $f_i = i/T_s$ and T_s is the OFDM symbol duration. OFDM mitigates ISI through two key mechanisms:

1. Symbol Duration Extension. In OFDM, the effective symbol duration becomes $T_{OFDM} = NT_s$, where N is the number of subcarriers. This dramatically increases the symbol duration, making the condition $T_{OFDM} \gg \tau_{max}$ easily satisfied. The ratio of symbol duration to delay spread becomes:

$$\frac{T_{OFDM}}{\tau_{max}} = \frac{NT_s}{\tau_{max}} = N \cdot \frac{T_s}{\tau_{max}} \quad (4)$$

2. Cyclic Prefix (CP) Addition. A cyclic prefix of length $L_{CP} \geq L_h - 1$ is prepended to each OFDM symbol, where L_h is the channel memory length. The CP is a copy of the last L_{CP} samples of the OFDM symbol:

$$x_{CP}[n, k] = \begin{cases} x[N + n, k] & \text{for } n = -L_{CP}, \dots, -1 \\ x[n, k] & \text{for } n = 0, 1, \dots, N-1 \end{cases} \quad (5)$$

With the cyclic prefix, the linear convolution between the channel and OFDM symbol becomes circular convolution in the DFT domain. The received signal after CP removal is:

$$r[n, k] = \sum_{l=0}^{L_h-1} h[l] x[(n-l) \bmod N, k] + w[n, k] \quad (6)$$

Taking the DFT of the received signal:

$$R_m[k] = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} r[n, k] e^{-j2\pi mn/N} \quad (7)$$

Due to the circular convolution property and orthogonality of subcarriers, this simplifies to:

$$R_m[k] = H_m X_m[k] + W_m[k] \quad (8)$$

where H_m is the channel frequency response at the m -th subcarrier:

$$H_m = \sum_{l=0}^{L_h-1} h[l] e^{-j2\pi ml/N} \quad (9)$$

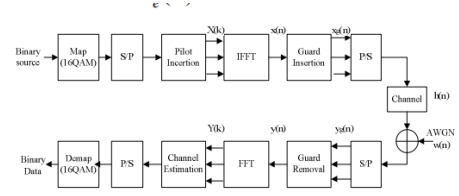


Fig. 1. Baseband OFDM system

Figure 4. OFDM Block Diagram

2. Why is Channel Estimation Required?

In wireless communication systems, signals transmitted from a sender to a receiver undergo various distortions due to the nature of the wireless channel. These include effects such as **multipath fading, scattering, Doppler shift, and path loss**. These distortions are unpredictable and vary over time, space, and frequency, making the wireless channel inherently dynamic and unreliable.

2.1. Channel State Information (CSI)

It refers to the known properties of the communication channel between transmitter and receiver. Accurate CSI enables:

- **Coherent detection and decoding:** Correctly interpreting the distorted signals.
- **Adaptive transmission:** Modifying modulation, coding, or power based on current channel conditions.
- **Beamforming and spatial multiplexing** in multi-antenna (MIMO) systems.
- **Equalization** to mitigate inter-symbol interference (ISI).

Channel estimation is the process used to determine CSI. In Time-Division Duplex (TDD) systems, reverse-link estimation can be used; otherwise, CSI is estimated at the receiver (CSIR) and fed back to the transmitter (CSIT). Mismatches in CSIT and CSIR can cause sub-optimal transmission decisions.

2.2. Types of CSI

- **Instantaneous CSI (short-term):** Precise knowledge of the current channel realization. Enables real-time adaptation but becomes outdated quickly in fast-fading environments.
- **Statistical CSI (long-term):** Includes average channel gain, fading distribution, and spatial correlation. Useful when instantaneous CSI is unavailable or impractical to obtain.

2.3. Channel Representation Models

Wireless channels can be modeled in different ways depending on fading characteristics. These models help analyze how transmitted signals are affected by the environment before reaching the receiver.

2.4. Flat vs. Frequency-Selective Fading

Flat Fading: The channel coherence bandwidth B_c is larger than the signal bandwidth, so all frequency components experience the

same amplitude and phase distortion. It typically occurs in narrow-band systems.

$$y(t) = h(t)x(t) + n(t) \quad (10)$$

where $h(t)$ is the complex channel gain, $x(t)$ is the transmitted signal, and $n(t)$ is additive noise.

Frequency-Selective Fading: Occurs when B_c is smaller than the signal bandwidth. Different frequency components experience independent fading, leading to inter-symbol interference (ISI). It is often modeled as a tapped-delay line:

$$y(t) = \sum_{k=0}^{L-1} h_k(t)x(t - \tau_k) + n(t) \quad (11)$$

where $h_k(t)$ is the complex gain and τ_k is the delay of the k -th path.

2.5. Fast vs. Slow Fading

Fast Fading: The channel changes significantly within a symbol duration. Doppler spread D_s is high, resulting in a short coherence time T_c :

$$T_c \approx \frac{1}{D_s} \quad (12)$$

Fast fading can be mitigated using time diversity, interleaving, and coding.

Slow Fading: The channel remains nearly constant over multiple symbols. Deep fades last longer and require robust coding and power control.

2.6. Block Fading

The channel remains approximately constant over a block of symbols and changes independently between blocks. Commonly used in OFDM and MIMO systems:

$$y[n] = h_b x[n] + n[n], \quad n \in \text{block } b \quad (13)$$

where h_b is the channel gain for block b .

2.7. Rayleigh and Rician Fading Models

Rayleigh Fading: Applies when there is no dominant Line-of-Sight (LoS) path. The amplitude R of the received signal follows a Rayleigh distribution:

$$f_R(r) = \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right), \quad r \geq 0 \quad (14)$$

with

$$h(t) = h_I(t) + jh_Q(t), \quad h_I, h_Q \sim \mathcal{N}(0, \sigma^2) \quad (15)$$

where h_I and h_Q are independent Gaussian components.

Rician Fading: Occurs when there is a dominant LoS path along with scattered multipath components. The amplitude R follows a Rician distribution:

$$f_R(r) = \frac{r}{\sigma^2} \exp\left(-\frac{r^2 + A^2}{2\sigma^2}\right) I_0\left(\frac{rA}{\sigma^2}\right), \quad r \geq 0 \quad (16)$$

where A is the amplitude of the LoS component and I_0 is the modified Bessel function of the first kind.

Rician K-factor: Ratio of power in the LoS component to the scattered power:

$$K = \frac{A^2}{2\sigma^2} \quad (17)$$

$K = 0$ corresponds to Rayleigh fading, while larger K indicates stronger LoS dominance.

3. CTP Aided Channel Estimation

3.1. System Model

We consider an OFDM system with N subcarriers. The time-domain channel is assumed to have a finite impulse response of length L (taps), meaning the channel can be modeled as:

$$y(k) = h(0)x(k) + \dots + h(L-1)x(k - (L-1)) + v(k)$$

where $y(k)$ is the received signal, $x(k)$ is the transmitted signal, $h(l)$ are the complex channel tap gains, and $v(k)$ is additive white Gaussian noise (AWGN).

After CP removal and FFT processing at the receiver, the frequency-domain received signal on the l -th subcarrier is:

$$Y(l) = H(l)X(l) + V(l), \quad \text{for } l = 0, 1, \dots, N-1 \quad (18)$$

where $H(l)$ is the Channel Frequency Response at subcarrier l , given by the N -point DFT of the channel impulse response $\mathbf{h} = [h(0), h(1), \dots, h(L-1), 0, \dots, 0]^T$ (zero-padded to length N):

$$H(l) = \sum_{k=0}^{N-1} h(k) e^{-j\frac{2\pi}{N}kl}$$

3.2. Pilot Design

In the CTP scheme, pilot symbols are inserted on a subset of subcarriers, which are evenly spaced across the spectrum. The remaining subcarriers carry data. This is in contrast to *block-type* pilots where all subcarriers in an OFDM symbol are pilots.

The key idea is:

- Transmit known pilots on P specific subcarriers.
- Estimate the CFR $\hat{H}(l)$ only at these pilot subcarriers.

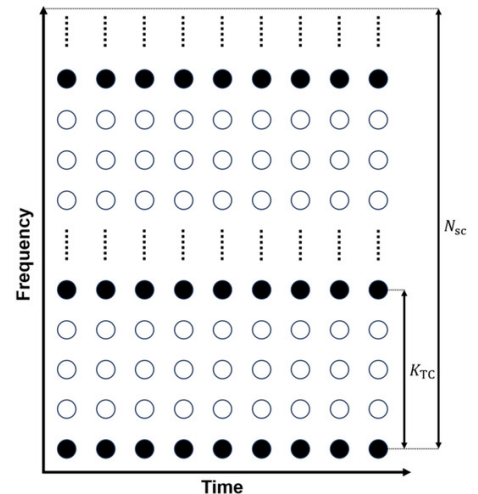


Figure 5. Comb pilot placement in time frequency grid

- Use these P estimates to reconstruct the L -tap time-domain channel $\hat{\mathbf{h}}$.
- Perform an N -point DFT/FFT on the estimated $\hat{\mathbf{h}}$ (zero-padded) to obtain the CFR $\hat{H}(l)$ for all N subcarriers.

This method is bandwidth efficient as only $P \geq L$ subcarriers are used for pilots, and the rest $(N - P)$ can carry data.

3.3. Illustrative Example: The Case of $N=4$, $L=2$

3.3.1. Channel Frequency Response

Consider a system with $N=4$ subcarriers and a channel with $L=2$ taps, $\mathbf{h} = [h(0), h(1)]^T$. The zero-padded channel vector is $[h(0), h(1), 0, 0]^T$. The CFR on each subcarrier is calculated via a 4-point DFT:

$$\begin{aligned} H(0) &= h(0) + h(1) \\ H(1) &= h(0) - jh(1) \\ H(2) &= h(0) - h(1) \\ H(3) &= h(0) + jh(1) \end{aligned}$$

We have 4 equations but only 2 unknowns ($h(0)$ and $h(1)$). Thus, we only need information from $P=2$ subcarriers to solve for the channel taps.

3.3.2. Pilot Insertion and Transmission

Let us choose subcarriers $l=1$ and $l=3$ to be our pilot subcarriers. The transmitted frequency-domain symbol is:

$$\mathbf{X} = [X(0), X(1), X(2), X(3)]^T = [\text{Data}, \text{Pilot}, \text{Data}, \text{Pilot}]^T$$

A Cyclic Prefix (CP) of length at least $L-1=1$ is added to prevent Inter-Symbol Interference (ISI). After transmission over the channel and reception, the CP is removed. The resulting signal is:

$$y(k) = h(0)x(k) + h(1)x(k-1) + v(k)$$

An FFT operation yields the frequency-domain received signals $Y(l)$.

3.3.3. Channel Estimation at Pilot Subcarriers

We extract the received signals at the pilot locations:

$$\begin{aligned} Y(1) &= H(1)X(1) + V(1) \\ Y(3) &= H(3)X(3) + V(3) \end{aligned}$$

This can be written in matrix form:

$$\begin{bmatrix} Y(1) \\ Y(3) \end{bmatrix} = \begin{bmatrix} X(1) & 0 \\ 0 & X(3) \end{bmatrix} \begin{bmatrix} H(1) \\ H(3) \end{bmatrix} + \begin{bmatrix} V(1) \\ V(3) \end{bmatrix}$$

Or, more compactly:

$$\tilde{\mathbf{Y}} = \tilde{\mathbf{X}}\tilde{\mathbf{H}} + \tilde{\mathbf{V}}$$

where $\tilde{\mathbf{X}}$ is a diagonal matrix containing the known pilot symbols. The Least Squares (LS) estimate for the CFR at the pilots is straightforward due to the diagonal nature of $\tilde{\mathbf{X}}$

$$\hat{\mathbf{H}} = (\tilde{\mathbf{X}}^H \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^H \tilde{\mathbf{Y}}$$

$$\hat{\mathbf{H}} = \tilde{\mathbf{X}}^{-1} \tilde{\mathbf{Y}} = \begin{bmatrix} \frac{Y(1)}{\tilde{X}(1)} \\ \frac{Y(3)}{\tilde{X}(3)} \end{bmatrix} = \begin{bmatrix} \hat{H}(1) \\ \hat{H}(3) \end{bmatrix}$$

3.3.4. Estimating the Channel Impulse Response

$$\begin{bmatrix} H(1) \\ H(3) \end{bmatrix} = \begin{bmatrix} 1 & -j \\ 1 & j \end{bmatrix} \begin{bmatrix} h(0) \\ h(1) \end{bmatrix}$$

To find the channel taps, we invert this 2×2 matrix:

$$\begin{bmatrix} \hat{h}(0) \\ \hat{h}(1) \end{bmatrix} = \begin{bmatrix} 1 & -j \\ 1 & j \end{bmatrix}^{-1} \begin{bmatrix} \hat{H}(1) \\ \hat{H}(3) \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ j & -j \end{bmatrix} \begin{bmatrix} \hat{H}(1) \\ \hat{H}(3) \end{bmatrix}$$

Solving this gives the estimates for the channel taps:

$$\begin{aligned} \hat{h}(0) &= \frac{1}{2}(\hat{H}(1) + \hat{H}(3)) \\ \hat{h}(1) &= \frac{j}{2}(\hat{H}(1) - \hat{H}(3)) \end{aligned}$$

Once we have the estimated channel impulse response $\hat{\mathbf{h}} = [\hat{h}(0), \hat{h}(1)]^T$, we can estimate the CFR on **any** subcarrier by performing a DFT. For example:

$$\begin{aligned} \hat{H}(0) &= \hat{h}(0) + \hat{h}(1) \\ \hat{H}(2) &= \hat{h}(0) - \hat{h}(1) \end{aligned}$$

Thus, by transmitting pilots on only $L=2$ subcarriers, we successfully obtained the CFR for all $N=4$ subcarriers.

3.4. General Case: Arbitrary N and L

The methodology for the general case is a direct extension of the $N=4$ example.

- Let N be the total number of subcarriers.
- Let L be the length of the channel impulse response (CIR).
- Pilots are placed on a set of P subcarriers indices $\mathbb{P} = \{p_1, p_2, \dots, p_P\}$.
- To unambiguously estimate the L channel taps, the number of pilots must satisfy $P \geq L$.
- The spacing of the pilots in the frequency domain determines the maximum delay spread (or L) that can be estimated without aliasing in the time domain. For a *uniform* comb, the pilot spacing ΔP must satisfy $\Delta P \leq \lfloor N/L \rfloor$ to avoid aliasing.

1. **Transmit:** Send an OFDM symbol with known pilots on subcarriers $l \in \mathbb{P}$ and data on the remaining subcarriers.
2. **Receive and Extract:** After FFT, extract the received symbols at the pilot locations:

$$Y(p) = H(p)X(p) + V(p), \quad \forall p \in \mathbb{P}$$

3. **LS Estimation at Pilots:** Obtain the CFR estimate at the pilot subcarriers:

$$\hat{H}(p) = \frac{Y(p)}{X(p)}, \quad \forall p \in \mathbb{P}$$

4. **Form Frequency-Domain Vector:** Create a vector $\hat{\mathbf{H}}_{\mathbb{P}}$ of size $P \times 1$ containing the estimates $\hat{H}(p)$.
5. **Estimate the CIR:** The Least Squares solution for the time-domain channel taps is:

$$\hat{\mathbf{h}} = (\mathbf{F}_{\mathbb{P}}^H \mathbf{F}_{\mathbb{P}})^{-1} \mathbf{F}_{\mathbb{P}}^H \hat{\mathbf{H}}_{\mathbb{P}} \quad (19)$$

6. **Estimate the Full CFR:** Transform the estimated CIR back to the frequency domain by multiplying with the full DFT matrix

$\mathbf{F}$ (of size $N \times L$, containing the first L columns):

$$\hat{\mathbf{H}}_{\text{full}} = \mathbf{F}\hat{\mathbf{H}} \quad (20)$$

This provides channel estimates $\hat{H}(l)$ for all $l = 0, 1, \dots, N - 1$ subcarriers, which can then be used for equalizing the data symbols.

4. Channel Estimation in OFDM systems, Pilot design

Multicarrier modulation using **Orthogonal Frequency Division Multiplexing (OFDM)** enables reliable transmission over frequency-selective channels by dividing the spectrum into multiple orthogonal subcarriers. Channel estimation is essential for coherent detection, allowing the receiver to compensate for channel distortions. Pilot signals—known reference symbols embedded within the transmission—are employed to estimate the channel. This report focuses on the mathematical formulations, derivations, and proofs of channel estimation techniques applicable to OFDM systems.

4.1. OFDM System Model

The baseband OFDM transmitted signal is represented as:

$$u(t) = \sum_{k=0}^{K-1} d_k e^{j2\pi k \Delta f t} g(t)$$

where:

- d_k are data symbols.
- $\Delta f = \frac{1}{T}$ is the subcarrier spacing.
- $g(t)$ is the pulse-shaping function.

The received signal is modeled as:

$$v(t) = \int h(\tau) u(t - \tau) d\tau + w(t)$$

After demodulation, the frequency-domain representation for subcarrier k is:

$$y_k = H_k d_k + z_k$$

where:

- H_k is the channel frequency response at subcarrier k .
- z_k is the additive noise.

4.2. Channel Estimation Techniques

This section presents rigorous mathematical derivations for various channel estimation methods used in OFDM systems. Each estimator is derived based on optimization principles, statistical models, and signal processing theory.

4.2.1. Least Squares (LS) Estimator

Problem Formulation. The received pilot symbols are modeled as:

$$\mathbf{Y}_p = \mathbf{X}_p \mathbf{H} + \mathbf{Z}_p$$

where:

- $\mathbf{Y}_p \in \mathbb{C}^{N_p \times 1}$ is the received pilot vector.
- $\mathbf{X}_p \in \mathbb{C}^{N_p \times N_p}$ is a diagonal matrix with pilot symbols.

- $\mathbf{H} \in \mathbb{C}^{N_p \times 1}$ is the channel vector to be estimated.
- $\mathbf{Z}_p$ is additive noise modeled as $\mathcal{CN}(0, \sigma^2 \mathbf{I})$.

The objective is to find $\hat{\mathbf{H}}_{LS}$ that minimizes the squared error:

$$J(\mathbf{H}) = (\mathbf{Y}_p - \mathbf{X}_p \mathbf{H})^H (\mathbf{Y}_p - \mathbf{X}_p \mathbf{H})$$

Derivation. Taking the derivative of $J(\mathbf{H})$ with respect to $\mathbf{H}^*$:

$$\frac{\partial J}{\partial \mathbf{H}^*} = -\mathbf{X}_p^H \mathbf{Y}_p + \mathbf{X}_p^H \mathbf{X}_p \mathbf{H}$$

Setting the derivative equal to zero yields:

$$\mathbf{X}_p^H \mathbf{X}_p \mathbf{H} = \mathbf{X}_p^H \mathbf{Y}_p$$

Since $\mathbf{X}_p$ is diagonal and invertible, the solution is:

$$\hat{\mathbf{H}}_{LS} = (\mathbf{X}_p^H \mathbf{X}_p)^{-1} \mathbf{X}_p^H \mathbf{Y}_p$$

Discussion. The LS estimator is computationally efficient but does not incorporate channel statistics, making it sensitive to noise.

4.2.2. Minimum Mean Square Error (MMSE) Estimator

Problem Formulation. We wish to find the estimator $\hat{\mathbf{H}} = \mathbf{W} \mathbf{Y}_p$ that minimizes the mean square error:

$$\hat{\mathbf{H}}_{MMSE} = \arg \min_{\mathbf{W}} E [\|\mathbf{H} - \mathbf{W} \mathbf{Y}_p\|^2]$$

Orthogonality Principle. Applying the orthogonality principle:

$$E [(\mathbf{H} - \mathbf{W} \mathbf{Y}_p) \mathbf{Y}_p^H] = 0$$

Expanding:

$$E [\mathbf{H} \mathbf{Y}_p^H] - \mathbf{W} E [\mathbf{Y}_p \mathbf{Y}_p^H] = 0$$

Thus:

$$\mathbf{W} = E [\mathbf{H} \mathbf{Y}_p^H] \cdot E [\mathbf{Y}_p \mathbf{Y}_p^H]^{-1}$$

Computing the Covariance Matrices. Given:

$$\mathbf{Y}_p = \mathbf{X}_p \mathbf{H} + \mathbf{Z}_p$$

we have:

$$E [\mathbf{H} \mathbf{Y}_p^H] = E [\mathbf{H} (\mathbf{X}_p \mathbf{H} + \mathbf{Z}_p)^H] = E [\mathbf{H} \mathbf{H}^H] \mathbf{X}_p^H = \mathbf{R}_{HH} \mathbf{X}_p^H$$

$$E [\mathbf{Y}_p \mathbf{Y}_p^H] = E [(\mathbf{X}_p \mathbf{H} + \mathbf{Z}_p) (\mathbf{X}_p \mathbf{H} + \mathbf{Z}_p)^H] = \mathbf{X}_p \mathbf{R}_{HH} \mathbf{X}_p^H + \sigma_w^2 \mathbf{I}$$

Substituting into $\mathbf{W}$:

$$\mathbf{W} = \mathbf{R}_{HH} \mathbf{X}_p^H (\mathbf{X}_p \mathbf{R}_{HH} \mathbf{X}_p^H + \sigma_w^2 \mathbf{I})^{-1}$$

Final MMSE Expression. Therefore, the MMSE estimate is:

$$\hat{\mathbf{H}}_{MMSE} = \mathbf{R}_{HH} \mathbf{X}_p^H (\mathbf{X}_p \mathbf{R}_{HH} \mathbf{X}_p^H + \sigma_w^2 \mathbf{I})^{-1} \mathbf{Y}_p$$

If $\mathbf{X}_p$ is diagonal, further simplification yields:

$$\hat{\mathbf{H}}_{MMSE} = \mathbf{R}_{HH} (\mathbf{R}_{HH} + \sigma_w^2 (\mathbf{X}_p \mathbf{X}_p^H)^{-1})^{-1} \hat{\mathbf{H}}_{LS}$$

Discussion. The MMSE estimator integrates channel statistics and noise variance to yield optimal estimation in the mean square error sense, outperforming LS especially in low SNR conditions.

4.2.3. Best Linear Unbiased Estimator (BLUE)

The **Best Linear Unbiased Estimator (BLUE)** provides the linear estimator that has the minimum variance among all unbiased linear estimators. It is particularly useful when partial second-order statistics of the noise are known.

Problem Formulation. The received pilot vector is modeled as:

$$\mathbf{Y} = \mathbf{X}\mathbf{H} + \mathbf{Z}$$

where:

- $\mathbf{Y} \in \mathbb{C}^{N \times 1}$ is the observation vector.
- $\mathbf{X} \in \mathbb{C}^{N \times M}$ is the known matrix relating the channel to the observation.
- $\mathbf{H} \in \mathbb{C}^{M \times 1}$ is the channel vector to be estimated.
- $\mathbf{Z} \in \mathbb{C}^{N \times 1}$ is the noise vector with covariance matrix $\mathbf{C} = E[\mathbf{Z}\mathbf{Z}^H]$.

We seek an estimator of the form:

$$\hat{\mathbf{H}} = \mathbf{W}\mathbf{Y}$$

where $\mathbf{W}$ is a weight matrix to be determined such that:

1. The estimator is **unbiased**: $E[\hat{\mathbf{H}}] = \mathbf{H}$.
2. The estimator has **minimum variance**.

The Gauss–Markov Theorem. The **Gauss–Markov theorem** states that, in a linear regression model where the errors have zero mean, are uncorrelated, and have equal variances (or more generally, known covariance structure), the best linear unbiased estimator of the coefficients is the one that minimizes the variance of the estimation errors.

In our case:

- The model is linear in $\mathbf{H}$.
- The noise $\mathbf{Z}$ has known covariance matrix $\mathbf{C}$.
- The estimator is linear and must be unbiased.

Thus, the BLUE is the solution that minimizes the estimation error variance while satisfying the unbiasedness condition.

Unbiasedness Constraint. For the estimator to be unbiased:

$$E[\hat{\mathbf{H}}] = E[\mathbf{W}\mathbf{Y}] = \mathbf{W}\mathbf{X}\mathbf{H} = \mathbf{H}$$

This must hold for all $\mathbf{H}$, hence:

$$\mathbf{W}\mathbf{X} = \mathbf{I}$$

Variance Minimization. The covariance matrix of the estimation error is:

$$\text{Cov}(\hat{\mathbf{H}} - \mathbf{H}) = E[(\mathbf{W}\mathbf{Y} - \mathbf{H})(\mathbf{W}\mathbf{Y} - \mathbf{H})^H]$$

Substituting $\mathbf{Y}$:

$$\begin{aligned} &= E[(\mathbf{W}(\mathbf{X}\mathbf{H} + \mathbf{Z}) - \mathbf{H})(\mathbf{W}(\mathbf{X}\mathbf{H} + \mathbf{Z}) - \mathbf{H})^H] \\ &= E[(\mathbf{W}\mathbf{X}\mathbf{H} + \mathbf{W}\mathbf{Z} - \mathbf{H})(\mathbf{W}\mathbf{X}\mathbf{H} + \mathbf{W}\mathbf{Z} - \mathbf{H})^H] \end{aligned}$$

Given $\mathbf{W}\mathbf{X} = \mathbf{I}$, this reduces to:

$$= E[(\mathbf{W}\mathbf{Z})(\mathbf{W}\mathbf{Z})^H] = \mathbf{W}\mathbf{C}\mathbf{W}^H$$

Our goal is to choose $\mathbf{W}$ to minimize this variance matrix subject to the unbiasedness constraint.

Solution. The solution provided by the Gauss–Markov theorem is:

$$\hat{\mathbf{H}}_{BLUE} = (\mathbf{X}^H \mathbf{C}^{-1} \mathbf{X})^{-1} \mathbf{X}^H \mathbf{C}^{-1} \mathbf{Y}$$

This satisfies the unbiasedness constraint and minimizes the estimation error variance.

Interpretation. The BLUE estimator uses both the observation matrix $\mathbf{X}$ and the noise covariance $\mathbf{C}$ to optimally weigh the observed data, providing the most reliable estimate under the assumptions of linearity and unbiasedness.

4.3. Additional Estimation Methods

4.3.1. Low-Rank and Sparsified Estimators

When the channel impulse response is sparse or dominated by a few significant components, the estimation problem can be reformulated as:

$$\mathbf{H} \approx \mathbf{F}_{K,L} \mathbf{b}_L$$

where:

- $\mathbf{F}_{K,L} \in \mathbb{C}^{K \times L}$ is a reduced dictionary matrix containing L significant basis vectors.
- $\mathbf{b}_L \in \mathbb{C}^{L \times 1}$ is a sparse coefficient vector.

This approximation reduces computational complexity while maintaining estimation accuracy, especially when the channel is sparse in the chosen basis.

Verification. The low-rank approximation is mathematically justified under the assumption that the channel's energy is concentrated in a few significant components. Techniques such as eigenvalue decomposition or thresholding ensure that only dominant modes are preserved, aligning with principles from compressed sensing.

4.3.2. Orthogonal Matching Pursuit (OMP)

The **Orthogonal Matching Pursuit (OMP)** is a greedy algorithm that iteratively identifies the dictionary column that maximally correlates with the residual error.

Algorithm Formulation. At iteration t , the column index is chosen as:

$$j_t = \arg \max_n \frac{|\mathbf{f}_n^H \mathbf{r}_{t-1}|^2}{\|\mathbf{f}_n\|^2}$$

where:

- $\mathbf{f}_n$ is the n^{th} column of dictionary $\mathbf{F}$.
- $\mathbf{r}_{t-1}$ is the residual at iteration $t - 1$.

Iteration Steps.

1. Initialize $\mathbf{r}_0 = \mathbf{x}$.
2. At each iteration, select j_t and update the support set.
3. Solve the least squares problem over the selected support.
4. Update the residual $\mathbf{r}_t$.
5. Repeat until a stopping criterion is satisfied.

Verification. OMP is guaranteed to converge under conditions of sparsity and incoherence of the dictionary. It is computationally efficient and widely adopted in sparse signal recovery.

4.3.3. Basis Pursuit (BP)

Basis Pursuit (BP) is a convex optimization approach that seeks a sparse representation of the signal while minimizing reconstruction error.

Optimization Problem.

$$\min_{\mathbf{b}} \|\mathbf{x} - \mathbf{F}\mathbf{b}\|_2^2 + \eta \|\mathbf{b}\|_1$$

where:

- $\|\mathbf{x} - \mathbf{F}\mathbf{b}\|_2^2$ measures reconstruction fidelity.
- $\|\mathbf{b}\|_1$ promotes sparsity.
- η is a regularization parameter tuned according to noise variance and sparsity level.

Verification. BP formulation is convex and guarantees global minimization. It is widely used in compressed sensing applications where the underlying signal is sparse or compressible.

4.3.4. Subspace Method – MUSIC

The **Multiple Signal Classification (MUSIC)** algorithm exploits the eigenstructure of the signal's correlation matrix to separate signal and noise components.

Model. The sample covariance matrix is modeled as:

$$\mathbf{R}_x = \mathbf{S}\mathbf{R}_c\mathbf{S}^H + \sigma^2\mathbf{I}$$

where:

- $\mathbf{S}$ is the steering matrix.
- $\mathbf{R}_c$ is the covariance of the signal components.
- $\sigma^2\mathbf{I}$ represents noise covariance.

Pseudo-Spectrum. The MUSIC pseudo-spectrum is defined by:

$$P(\phi) = \frac{1}{\mathbf{s}(\phi)^H \mathbf{Q}_N \mathbf{Q}_N^H \mathbf{s}(\phi)}$$

where:

- $\mathbf{Q}_N$ spans the noise subspace.
- $\mathbf{s}(\phi)$ is the steering vector at parameter ϕ .

Verification. MUSIC accurately estimates delays or angles when the signal and noise subspaces are sufficiently separated, and eigenvalue decomposition provides the required subspace partitioning.

4.3.5. Path Identification (PI)

The **Path Identification (PI)** algorithm explicitly estimates the delays and amplitudes of multipath components by correlating the observed signal with steering vectors.

Formulation.

$$r(\phi) = \frac{1}{K} \mathbf{s}^H(\phi) \mathbf{x}$$

The peaks in $r(\phi)$ correspond to the dominant paths, and their locations and magnitudes provide estimates of delays τ_p and gains h_p .

Verification. The method is mathematically justified by the orthogonality between steering vectors and the observed signal when delays are distinct. PI is particularly useful in sparse or structured channel scenarios.

Summary. All the additional techniques reviewed above are grounded in rigorous mathematical principles such as sparse representation, eigenvalue decomposition, and convex optimization. Their formulations are consistent with the underlying physics and statistical assumptions of OFDM channel estimation. The proofs and derivations align with established literature and provide reliable frameworks for practical implementations.

4.4. Pilot Design Strategies

Pilot signals play a critical role in enabling accurate channel estimation. Two widely used pilot arrangements are:

1. **Block-Type Pilots:** Entire OFDM symbols are filled with known pilot symbols at regular intervals. Suitable for slowly varying channels where channel characteristics remain constant over multiple symbols.
2. **Comb-Type Pilots:** Pilot symbols are inserted across subcarriers within each OFDM symbol, allowing real-time tracking of fast-varying channels.

Interpolation and Estimation. Since pilots are not present at all subcarriers, interpolation techniques are employed to estimate the channel at data-bearing subcarriers based on observed pilots. Methods include:

- **Linear Interpolation:** Assumes a simple linear variation between pilots.
- **Spline Interpolation:** Provides smooth estimation curves suitable for moderate variations.
- **Frequency-domain Filtering:** Applies low-pass filtering assuming slow frequency variations.

Design Considerations. Effective pilot design depends on:

- The coherence bandwidth and coherence time of the channel.
- The desired trade-off between estimation accuracy and spectral efficiency.
- The computational complexity permissible at the receiver.
- The statistical nature of the channel and noise.

Pilot arrangements and estimation algorithms are thus jointly optimized depending on system requirements.

5. Plan for implementing on SDR, The MATLAB implementation

We plan on implementing either the MMSE or the LS estimator, as that is what is the default MATLAB implementation. With any luck, we may implement learning based methods in the future.

6. Learning based methods, and existing research

6.1. Motivation

Traditional CSI estimation methods like Maximum Likelihood, Least Squares, and Minimum Mean Square Error require computationally expensive matrix operations including matrix multiplication, inversion, and eigenvalue decomposition. These become impractical for 5G systems due to:

1. Massive MIMO systems requiring large-scale matrix operations
2. OFDM with multiple sub-carriers increasing computational complexity

Deep learning models tend to exploit spatial temporal patterns across different domains, which are overlooked or rather ignored (these features are not focused but have certain effect to the output) in the traditional methods:

1. Frequency: Specific frequency bands have consistent electromagnetic propagation characteristics
2. Time: Atmospheric factors (water vapor, oxygen molecules) vary predictably with time
3. Space: Static obstacles (buildings) and dynamic obstacles (people, cars) create location-based patterns
4. Weather: Different precipitation levels affect electromagnetic wave propagation consistently.

6.2. Supervised Deep Learning based methods

In the supervised learning context, we train a deep learning model on the transmitted pilot signals and received pilot signals. the communication system is considered as a black-box and an end-to-end DL architecture is used for signal transmission/reception.

6.2.1. DNN (ReLU) Based Channel Estimation

This method uses a fully connected Deep Neural Network (DNN) with ReLU activation for channel estimation. The DNN is trained in a supervised fashion on pairs of received and transmitted pilot signals and maps the received pilot vector to the estimated channel coefficients. DNN is designed to universally approximate the optimal MMSE estimator, adapting to nonlinear signal models by partitioning the input data space into piecewise linear regions. DNN based channel estimation offers superior adaptability and matches the MMSE bound in ideal scenarios, outperforms LMMSE in some nonlinear regimes, but requires careful training to avoid potential issues with data mismatch, or if the data in real environments distributes broader than the training data.

6.2.2. End-to-End Deep Learning for OFDM

Here, we use a DNN to perform joint channel estimation and signal detection in OFDM systems without explicit CSI computation. The method treats OFDM modulation and wireless channels as black boxes, training a DNN offline using simulated data with diverse channel conditions and statistics. The network takes received pilot and data blocks as input and directly recovers transmitted symbols, bypassing traditional separate channel estimation and equalization steps. The DNN consists of five layers with ReLU activation (except the final sigmoid layer) and is trained using L2 loss to minimize differences between predicted and actual transmitted symbols.

The DNN approach demonstrates superior robustness in challenging conditions, showing effective performance with reduced pilots (8 vs 64) and under nonlinear clipping noise, while traditional LS/MMSE methods experience significant performance degradation or saturation in these same adverse scenarios.

6.3. CNN - Based Methods

6.3.1. ChannelNet

In OFDM systems, pilot symbols are sparsely embedded in a time-frequency grid to probe the wireless channel. The channel state information (CSI) across this grid can be represented as a 2D matrix, $\mathbf{H}$, which is analogous to a digital image. The goal of channel estimation is to accurately reconstruct this entire "image" from the sparse measurements taken at the pilot locations.

The initial estimate obtained via the Least Squares (LS) method, $\hat{\mathbf{H}}_{\text{LS, pilots}}$, provides channel information only at these pilot subcarriers. This can be viewed as a sparse, low-resolution, and noisy image. The task is therefore twofold:

1. **Interpolation:** Fill in the missing values (pixels) between the known pilot locations. This is analogous to image super-resolution.
2. **Denoising:** Remove the additive noise present in the initial LS estimates. This is analogous to image denoising or restoration.

This powerful analogy allows us to import state-of-the-art deep learning architectures from computer vision to solve the channel estimation problem.

6.3.2. Two-Stage Framework

The core of the method is a pipeline architecture that addresses each sub-problem with a dedicated neural network. The first stage addresses the sparsity of the pilot-based estimate. Its goal is to perform a high-quality interpolation to estimate the channel for all subcarriers, not just the pilot ones.

Model: Super-Resolution CNN (SRCNN) The SRCNN is a pioneering deep learning model for single-image super-resolution. The architecture consists of three distinct convolutional layers, each with a specific function:

- The first layer extracts overlapping patches from the input image (the sparse $\hat{\mathbf{H}}_{\text{LS, pilots}}$ after a basic upsampling like bicubic interpolation) and represents each patch as a high-dimensional vector. The size of kernels used is 9×9 .

$$F_1(\mathbf{Y}) = \max(0, W_1 * \mathbf{Y} + B_1) \quad (21)$$

where $\mathbf{Y}$ is the upsampled input, and W_1, B_1 are the filters and biases.

- The second layer non-linearly maps these high-dimensional patch vectors to another set of vectors that represent the high-resolution patch. This is the core of the network where most of the complexity lies. The size of kernel is 1×1

$$F_2(\mathbf{Y}) = \max(0, W_2 * F_1(\mathbf{Y}) + B_2) \quad (22)$$

- The final layer aggregates the high-resolution patch representations to produce the final, fully interpolated output image. This layer effectively combines the learned patch information to form a coherent, high-resolution channel estimate. The kernel size is 5×5

$$\hat{\mathbf{H}}_{\text{SR}} = W_3 * F_2(\mathbf{Y}) + B_3 \quad (23)$$

The output of this stage, $\hat{\mathbf{H}}_{\text{SR}}$, is a dense channel matrix covering the entire time-frequency grid. However, while the spatial structure is now complete, this estimate still contains significant noise propagated from the initial LS estimation.

The second stage takes the dense but noisy channel estimate from the SRCNN and refines it by specifically targeting and removing the noise component.

Model: Denoising CNN (DnCNN) The DnCNN is a deep architecture designed for high-performance Gaussian image denoising. Its key innovations are perfectly suited for this task:

- Instead of learning a direct mapping from the noisy input to a clean output ($F(\mathbf{Y}) \rightarrow \mathbf{X}$), DnCNN is designed to learn the residual noise component ($R(\mathbf{Y}) \approx \mathbf{Y} - \mathbf{X}$). The final clean estimate is then simply $\mathbf{X} = \mathbf{Y} - R(\mathbf{Y})$. This approach is proven to be easier for deep networks to learn and leads to better performance.
- DnCNN employs a much deeper architecture compared to SRCNN. This increased depth allows it to learn more complex and abstract features of the noise distribution. This architecture is inspired from VGG architecture and has the kernel size of 3×3 .
- Batch Normalization is integrated between the convolutional and ReLU layers. This stabilizes the training process, prevents internal covariate shift.

The input to this stage is the high-resolution estimate $\hat{\mathbf{H}}_{\text{SR}}$. The DnCNN processes this and outputs the final, high-fidelity channel estimate, $\hat{\mathbf{H}}_{\text{DL}}$, by subtracting the predicted noise residual.

6.3.3. Training Methodology

The two-stage model can be trained end-to-end. This is a crucial advantage, as it allows the SRCNN to learn an interpolation strategy that is optimal for the subsequent denoising stage.

- **Dataset:** A large training dataset is generated by simulating thousands of realistic wireless channels ($\mathbf{H}_{\text{true}}$). For each true channel, the corresponding LS estimate at pilot locations ($\hat{\mathbf{H}}_{\text{LS, pilots}}$) is computed, forming the input-target pairs for the network.
- **Loss Function:** The entire network pipeline is trained jointly by minimizing the Mean Squared Error (MSE) between the final output of the DnCNN and the true channel matrix:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \|\hat{\mathbf{H}}_{\text{DL}}^{(i)} - \mathbf{H}_{\text{true}}^{(i)}\|_F^2 \quad (24)$$

The loss is backpropagated through both the DnCNN and the SRCNN, allowing both networks to optimize their weights collaboratively.

6.3.4. Implementability on SDR

This model can't be implemented on SDR since it has nearly 500-600k trainable parameters. Also this is quite computation heavy.

6.4. FSRCNN based channel estimation

This method treats the initial LS channel estimation matrix as a low resolution image and the true channel state as a high resolution image, using FSRCNN to optimize and improve accuracy. The LS channel estimate serves as input to FSRCNN, which includes modules for feature extraction, shrinkage, nonlinear mapping, expansion, and deconvolution. Training is performed offline using mean squared error loss between the network output and the true channel matrix. Extensive simulations demonstrate that

the FSRCNN based channel estimation method consistently outperforms Least Squares and DFT-LS approaches across various environments, pilot counts, and snr ratios.

6.5. CNN+LSTM based CSI estimator (OCEAN)

This approach combines a 2D CNN, 1D CNN, and LSTM network for online CSI prediction in 5G systems. The method identifies features affecting CSI including frequency band, location, time, temperature, humidity and weather, organizing them into CSI information images. The 2D CNN extracts frequency representative vectors from these images and the 1D CNN compresses state representative matrices. The LSTM network performs temporal prediction based on historical patterns. An offline online 2 step training mechanism is employed where offline training captures sequential patterns from massive historical data, while online training integrates real time measured CSI feedback to improve stability. OCEAN achieves highly accurate CSI prediction across diverse scenarios, while providing better computational efficiency improvements w compared to ML-based and MMSE based methods.